The algebraic small object argument as a saturating operation

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joint work in progress with Christian Sattler

Weak factorization systems

⊗ On a category $\mathscr{E}$, pair $(\mathcal{L}, \mathcal{R})$ of $\mathcal{L}, \mathcal{R} \subseteq \mathrm{Ob}\,\mathscr{E}^{\to}$

$$\begin{tikzcd} & \exists \arrow[dr, dashed, "\in \mathcal{R}"] & \\ \circ \arrow[ur, dashed, "\mathcal{L} \ni"] \arrow[rr] & & \circ \end{tikzcd} \qquad \begin{tikzcd} \circ \arrow[r] \arrow[d, "\mathcal{L} \ni"'] & \circ \arrow[d, "\in \mathcal{R}"] \\ \circ \arrow[r] \arrow[ur, dashed, "\exists"] & \circ \end{tikzcd}$$

⊗ Often generated by a set $\mathcal{S} \subseteq \mathrm{Ob}\,\mathscr{E}^{\to}$ of left maps

$$\mathcal{R} = \{\text{maps right lifting against all } f \in \mathcal{S}\}$$

⊗ (complemented mono, split epi) in **Set** generated by $\{0 \rightarrowtail 1\}$

⊗ (triv cofib, Kan fib) on $\mathrm{PSh}(\Delta)$ generated by $\{\Lambda^n_k \rightarrowtail \Delta^n \mid k \leq n \in \mathbb{N}\}$

Generation by a category

- f right lifts against $u : \mathscr{J} \to \mathscr{E}^{\to}$ when

$$\forall j \in \mathscr{J} \qquad\qquad \forall \alpha : i \to j$$

$$\begin{tikzcd} \circ \arrow[r] \arrow[d, "u_j"'] & \circ \arrow[d, "f"] \\ \circ \arrow[r] \arrow[ur, dashed] & \circ \end{tikzcd} \qquad \begin{tikzcd} \circ \arrow[r] \arrow[d] \arrow[dr, phantom, "u_\alpha"] & \circ \arrow[r] \arrow[d] & \circ \arrow[d, "f"] \\ \circ \arrow[r] \arrow[urr, dashed] & \circ \arrow[r] \arrow[ur, dashed] & \circ \end{tikzcd}$$

- (complemented mono, split epi) in **AbGrp** generated by full subcategory of $\mathscr{E}^{\to}$ containing

$$\begin{tikzcd} 0 \arrow[d, tail] \\ \mathbb{Z} \end{tikzcd} \qquad \begin{tikzcd} \mathbb{Z} \arrow[d, tail, "\Delta"] \\ \mathbb{Z} \times \mathbb{Z} \end{tikzcd}$$

- "uniform fibrations" in, e.g., cubical sets (used in models of HoTT)

Quillen's small object argument

⊗ Build a WFS from a set $\mathcal{S}$ of generators (assuming …)

$$\begin{tikzcd}
 & \coprod \Lambda^n_k \arrow[r, tail] \arrow[dl] & \coprod \Delta^n \arrow[d] & \coprod \Lambda^n_k \arrow[r, tail] \arrow[dl] & \coprod \Delta^n \arrow[d] & & \\
X \arrow[rr, tail] \arrow[d] & & X_1 \arrow[ul, phantom, very near start, "\ulcorner"] \arrow[rr, tail] \arrow[dll] & & X_2 \arrow[ul, phantom, very near start, "\ulcorner"] \arrow[dllll] & \cdots & X_\kappa \arrow[dllllll, -open triangle] \\
Y & & & & & &
\end{tikzcd}$$

⊗ The left factor is a transfinite composite of cobase changes of coproducts of generating maps

⊗ Any left map is a retract (in $\mathscr{E}^{\to}$) of one of these

Saturation

Def. $\mathcal{A} \subseteq \operatorname{Ob} \mathscr{E}^{\to}$ is saturated when it is closed under coproducts, cobase change, transfinite composition, and retracts.

- When $(\mathcal{L}, \mathcal{R})$ is generated from $\mathcal{S}$ by the SOA, $\mathcal{L}$ is the least saturated class containing $\mathcal{S}$.

Saturation for categories?

- Garner's algebraic small object argument (2008) generates a WFS given $u\colon \mathscr{J} \to \mathscr{E}^{\to}$
- Builds left factors as transfinite composites

$$X \longrightarrow X_1 \longrightarrow X_2 \longrightarrow \cdots \longrightarrow X_\kappa$$

... but step maps may not be left maps!

- 💡: see

$$\begin{tikzcd}
X \arrow[r, equal] \arrow[d] & X \arrow[r, equal] \arrow[d] & X \arrow[r, equal] \arrow[d] & \cdots \arrow[r, equal] & X \arrow[d] \\
X \arrow[r] & X_1 \arrow[r] & X_2 \arrow[r] & \cdots \arrow[r] & X_\kappa
\end{tikzcd}$$

as a colimit in the category **LeftMap** of left-structured maps

Result

⊗ Left factors are built by colimits of diagrams in **LeftMap**:

(a) sequential colimits

$$\begin{tikzcd}
\circ \arrow[r] \arrow[d, tail] & \circ \arrow[r] \arrow[d, tail] & \circ \arrow[r] \arrow[d, tail] & \cdots \\
\circ \arrow[r] & \circ \arrow[r] & \circ \arrow[r] & \cdots
\end{tikzcd}$$

(b) pushouts

$$\begin{tikzcd}
& \circ \arrow[r] \arrow[dl] \arrow[dd, tail] & \circ \arrow[dd, tail] \\
\circ \arrow[dd, tail] & & \\
& \circ \arrow[r] \arrow[dl] & \circ \\
\circ & &
\end{tikzcd}$$

(c) colimits of diagrams factoring through $\mathscr{J}$

$$\mathscr{I} \xrightarrow{d} \mathscr{J} \xrightarrow{u} \mathbf{LeftMap}$$

⊗ and "vertical" composition $\circ \rightarrowtail \circ \rightarrowtail \circ$

Wrap-up

- Can parlay idea into universal property of the double category of left-structured maps

- Under conditions on $u\colon \mathscr{J} \to \mathscr{E}^{\to}$, form of colimits can be further constrained, *cf.* Athorne's coalgebraic cell complexes (2014)

- Saturation principle for WFS generated by a set is a special case

POST-TALK UPDATE:
This slide was too optimistic. We:
- prove a partial universal property, giving existence but not uniqueness
- do not know if we can recover the standard principle for a WFS generated by a set as a special case.

For more details, see the preprint: https://arxiv.org/abs/2506.02759

Thank you!

References

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